NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83401/24
6 December 1994

PERFORMANCE SPECIFICATION

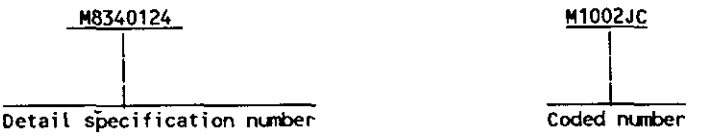
RESISTOR NETWORK, FIXED, FILM,
STYLE RZ240, 10 PIN SIP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RZ240, 10 pin single-in-line package (SIP) resistor networks with longer leads. This style is available in all characteristics and resistance tolerances.

1.2 Part or Identifying Number (PIN). Resistor networks covered by this specification are identified by a PIN which should consist of the basic number of this specification and a coded number. The PIN shall be in the following form:



The coded number shall be derived in accordance with MIL-R-83401.

1.2.1 Schematics. The schematic of the resistor network shall be identified by a single letter in accordance with the drawings below. The resistor element R_{REF} shall be the reference element used in determining the ratio accuracy (when applicable).

1.2.1.1 Standard resistance values. The standard resistance values and the resistance value designators for the H schematic are as specified in table I.

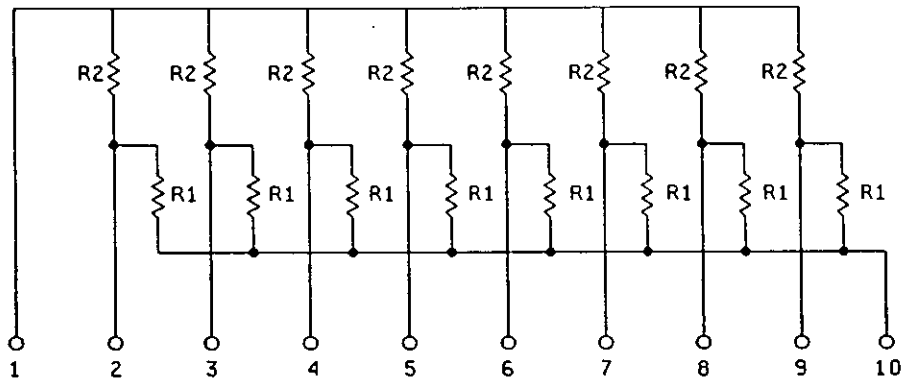
TABLE I. Standard resistance values.

Resistance designator	R_1 (ohms)	R_2 (ohms)		Resistance designator	R_1 (ohms)	R_2 (ohms)
A001	82	130		A011	330	680
A002	120	200		A012	1.5k	3.3k
A003	130	210		A013	3.0k	6.2k
A004	160	260		A014	180	270
A005	180	240		A015	270	270
A006	180	390		A016	560	560
A007	220	270		A017	560	1.2k
A008	220	330		A018	620	2.7k
A009	330	390		A019	150	1.0k
A010	330	470		A020	1.0k	1.0k

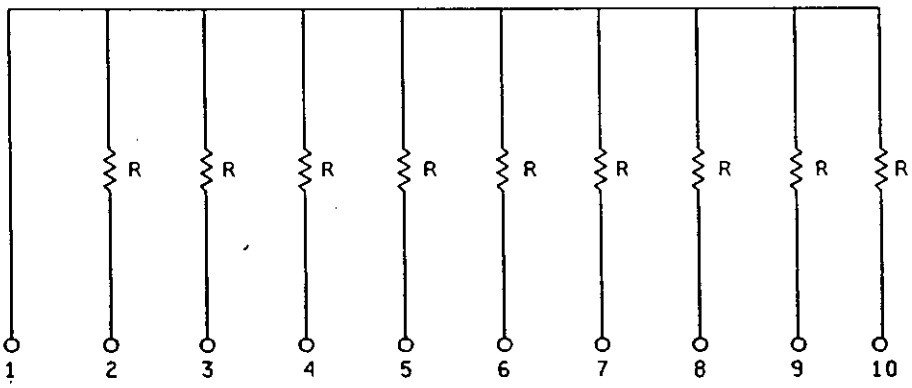
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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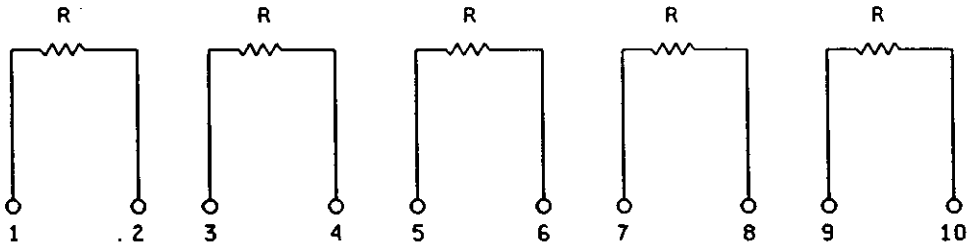
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SCHEMATIC "H"



SCHEMATIC "C"



SCHEMATIC "G"

FIGURE 1. Schematics.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-R-39032 - Resistors, Packaging of.
MIL-R-83401 - Resistor Networks, Fixed, Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed Resistors, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-83401, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 2.

3.3 Power rating (70°C). The power ratings for schematics for C, G, and H shall be as specified in table II.

TABLE II. Power ratings.

Schematics	Characteristic	Element power rating (watts)	Network power rating (watts)
C	C	0.06	0.54
	V	0.06	0.54
	H	0.12	1.08
	K	0.12	1.08
	M	0.12	1.08
G	C	0.06	0.30
	V	0.06	0.30
	H	0.12	0.60
	K	0.12	0.60
	M	0.12	0.60
H	C	0.04	0.54
	V	0.04	0.54
	H	0.07	1.08
	K	0.07	1.08
	M	0.07	1.08

3.4 Power conditioning. The power applied for power conditioning for schematics C, G, and H shall be 1.5 times rated power.

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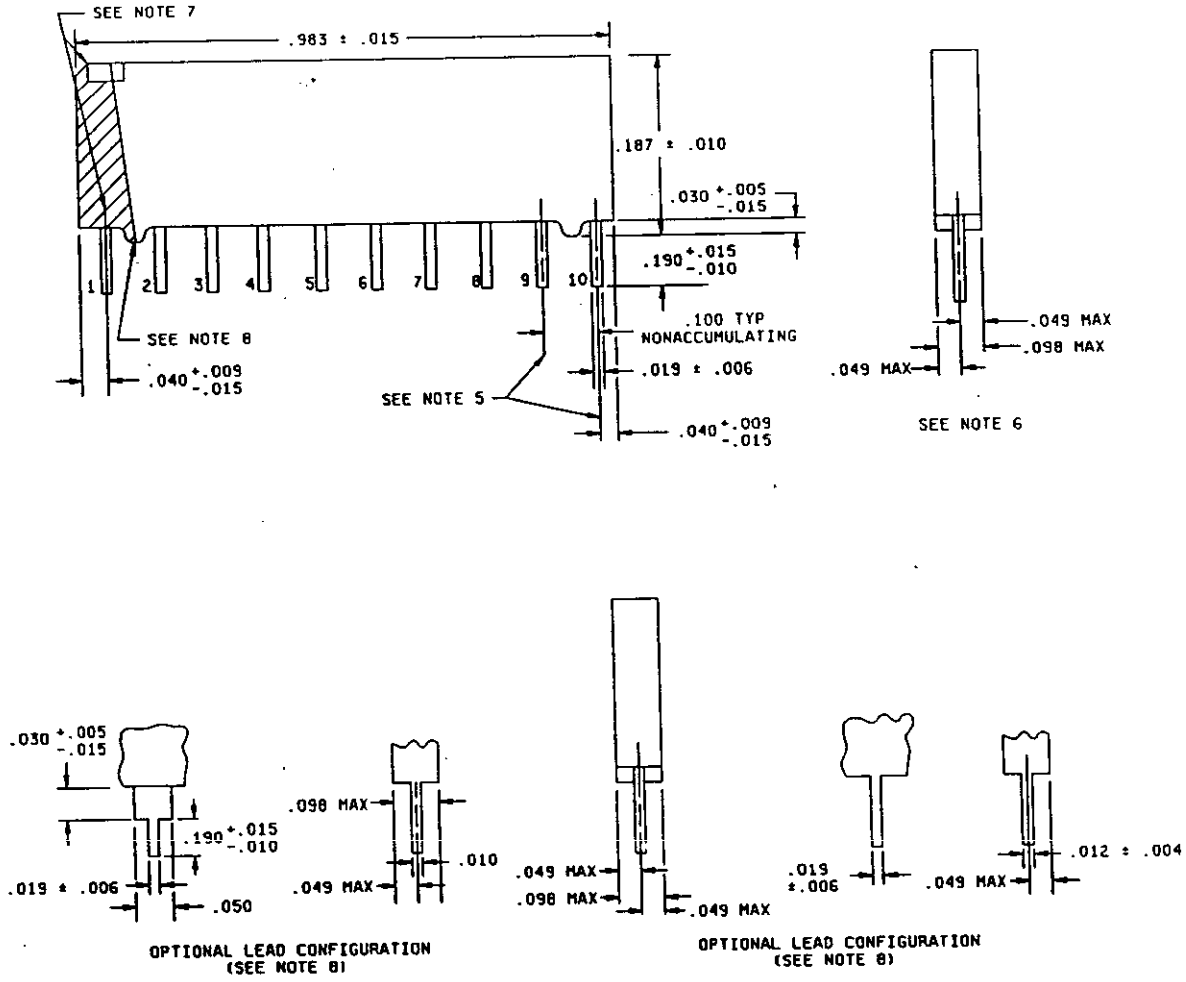


FIGURE 2. Style RZ240 resistors.

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Inches	mm	Inches	mm
.003	0.08	.024	0.61
.004	0.10	.030	0.76
.005	0.13	.040	1.02
.009	0.23	.049	1.24
.010	0.25	.050	1.27
.012	0.30	.074	1.88
.014	0.36	.089	2.26
.015	0.38	.100	2.54
.017	0.43	.187	4.75
.020	0.51	.190	4.83
		.983	24.97

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. The picturization of the styles above is given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope and do not alter the functional aspects of the device are acceptable.
5. Terminal centerline to centerline measurements made at point of emergence of the lead from the body.
6. Measurement made at point of emergence of the lead from the body.
7. Pin 1 locator shall be a bevel, bar or a dot above pin number 1 in the shaded area.
8. If the standoffs are located on the body, a minimum of two standoffs are required as illustrated. As an option, additional standoffs may be located on the body of the resistor network. If leads with standoffs are used, standoffs on the body are not required.

FIGURE 2. Style RZ240 resistors - Continued.

3.5 Voltage rating. The maximum continuous working voltage for each resistor shall not exceed 50 volts dc or ac rms.

3.6 Resistance. Minimum and maximum resistance values shall be as follows:

Resistance tolerance	Minimum resistance (ohms)	Maximum resistance (megohms)
B (±0.1 percent)	100	0.1
D (±0.5 percent)	100	0.1
F (±1.0 percent)	10.0	1.0
G (±2.0 percent)	10.0	1.0
J (±5.0 percent)	10.0	1.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-83401.

4.2 H schematic tests. For the test on H schematic, the dc resistance measurements, resistance temperature characteristic measurement, and short time overload power application shall be made between each terminal and the applicable common terminal without compensation for the shunt circuits. The pin to pin dc resistance measurements shall be used directly to determine the acceptability to the requirements herein.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-R-83401 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
DLA - ES

(Project 5905-1407)

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-83401/24	2. DOCUMENT DATE (YYMMDD) 94-12-06

3. DOCUMENT TITLE

RESISTOR NETWORK, FIXED, FILM, STYLE RZ240, 10 PIN SIP

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commander Defense Electronics Supply Center ATTN: DESC-ELDM	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-5257 (2) AUTOVON DSN 986-5257 FAX 296-8868
c. ADDRESS (Include Zip Code) 1507 Wilmington Pike Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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